

CLASSICAL MECHANICS: MASTER PROOFS MANUAL

Complete 42-Theorem Rigorous Derivation & Diagram Handbook

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Complete Standalone Solution Manual with TikZ Diagrams for Every Proof (sciqb.com)

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Chapter 1

Laws of Vectors in 2D & System of Coplanar Forces

1.1 Parallelogram Law of Forces

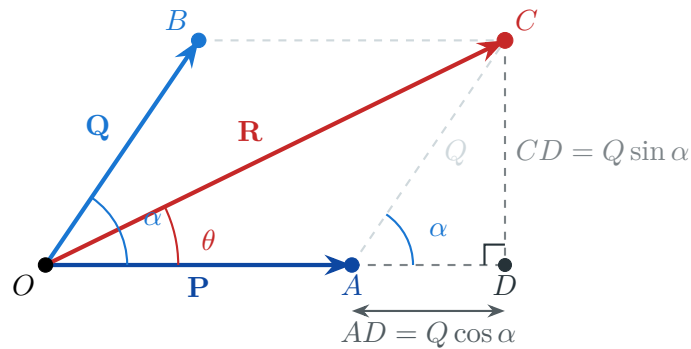
Theorem / Law: Proof 1: Parallelogram Law of Forces

If two coplanar forces **P** and **Q** acting simultaneously on a particle at point *O* at angle α are represented in magnitude and direction by two adjacent sides of a parallelogram, their resultant **R** is represented in magnitude and direction by the diagonal passing through *O*:

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}, \quad \tan \theta = \frac{Q \sin \alpha}{P + Q \cos \alpha} \quad (1.1)$$

Rigorous Proof / Derivation: Derivation of Parallelogram Law

Let *OA* represent force *P* along *X*-axis and *OB* represent force *Q* at angle α to *OA*. Complete parallelogram *OACB*. Draw perpendicular *CD* onto *OA* produced.



In right-angled triangle $\triangle ADC$: $AD = Q \cos \alpha$, $CD = Q \sin \alpha$. In right-angled triangle $\triangle ODC$:

$$OC^2 = OD^2 + CD^2 = (OA + AD)^2 + CD^2 = (P + Q \cos \alpha)^2 + (Q \sin \alpha)^2$$

$$R^2 = P^2 + 2PQ \cos \alpha + Q^2(\cos^2 \alpha + \sin^2 \alpha) = P^2 + Q^2 + 2PQ \cos \alpha \implies R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}$$

$$\tan \theta = \frac{CD}{OD} = \frac{Q \sin \alpha}{P + Q \cos \alpha}$$

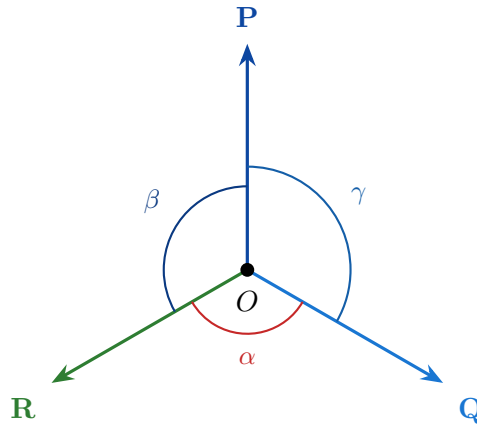
1.2 Lami's Theorem

Theorem / Law: Proof 2: Lami's Theorem

If three coplanar concurrent forces $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{R}$ acting at a point O are in static equilibrium, each force is proportional to the sine of the angle between the other two forces:

$$\frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma} \quad (1.2)$$

Rigorous Proof / Derivation: Derivation of Lami's Theorem



Since forces $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{R}$ are in static equilibrium:

$$\mathbf{P} + \mathbf{Q} + \mathbf{R} = \mathbf{0} \implies \mathbf{P} + \mathbf{Q} = -\mathbf{R}$$

Taking vector cross product with $\mathbf{P}$:

$$\mathbf{P} \times (\mathbf{P} + \mathbf{Q} + \mathbf{R}) = \mathbf{0} \implies \mathbf{P} \times \mathbf{Q} = \mathbf{R} \times \mathbf{P}$$

Similarly, taking vector cross product with $\mathbf{Q}$:

$$\mathbf{Q} \times \mathbf{R} = \mathbf{P} \times \mathbf{Q}$$

Evaluating vector magnitudes:

$$PQ \sin(180^\circ - \gamma) = QR \sin(180^\circ - \alpha) = RP \sin(180^\circ - \beta)$$

Dividing by PQR :

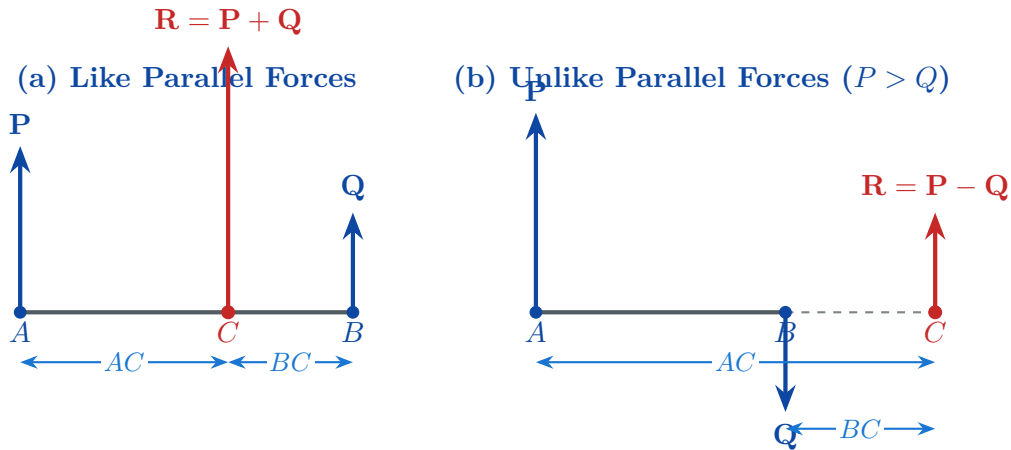
$$\frac{\sin \gamma}{R} = \frac{\sin \alpha}{P} = \frac{\sin \beta}{Q} \implies \frac{P}{\sin \alpha} = \frac{Q}{\sin \beta} = \frac{R}{\sin \gamma}$$

1.3 System of Parallel Forces

Theorem / Law: Proof 3 & 4: Resultant of Like and Unlike Parallel Forces

1. **Like Parallel Forces P and Q :** Resultant $R = P + Q$ acting parallel to P and Q at internal point C ($P \cdot AC = Q \cdot BC$). 2. **Unlike Parallel Forces $P > Q$:** Resultant $R = P - Q$ acting parallel to P at external point C ($P \cdot AC = Q \cdot BC$).

Rigorous Proof / Derivation: Derivation of Parallel Forces Resultants



Taking moments about base point A:

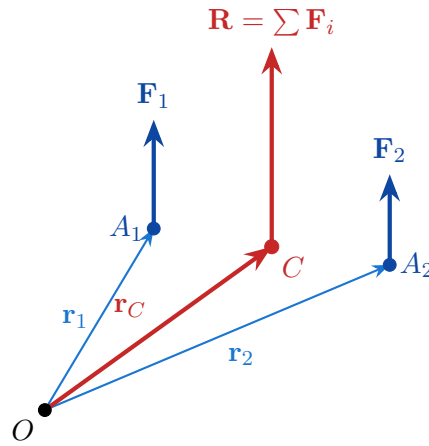
$$R \cdot AC = Q \cdot AB \implies (P \pm Q) \cdot AC = Q \cdot (AC \pm BC) \implies P \cdot AC = Q \cdot BC \implies \frac{AC}{BC} = \frac{Q}{P}$$

Theorem / Law: Proof 5: Center of Parallel Forces

The position vector $\mathbf{r}_C$ of the center of a system of parallel forces $\mathbf{F}_i = F_i \hat{u}$ acting at points $\mathbf{r}_i$ is independent of the direction $\hat{u}$:

$$\mathbf{r}_C = \frac{\sum_{i=1}^n F_i \mathbf{r}_i}{\sum_{i=1}^n F_i} \quad (1.3)$$

Rigorous Proof / Derivation: Derivation of Center of Parallel Forces



Total vector moment about origin O :

$$\mathbf{G}_O = \sum_{i=1}^n (\mathbf{r}_i \times \mathbf{F}_i) = \sum_{i=1}^n (\mathbf{r}_i \times F_i \hat{u}) = \left(\sum_{i=1}^n F_i \mathbf{r}_i \right) \times \hat{u}$$

Equating to moment of single resultant force $\mathbf{R} = (\sum F_i) \hat{u}$ acting at $\mathbf{r}_C$:

$$\mathbf{r}_C \times \mathbf{R} = \mathbf{r}_C \times \left(\sum F_i \hat{u} \right) = \left(\mathbf{r}_C \sum F_i \right) \times \hat{u}$$

Comparing coefficients yields $\mathbf{r}_C \sum F_i = \sum F_i \mathbf{r}_i \implies \mathbf{r}_C = \frac{\sum F_i \mathbf{r}_i}{\sum F_i}$.

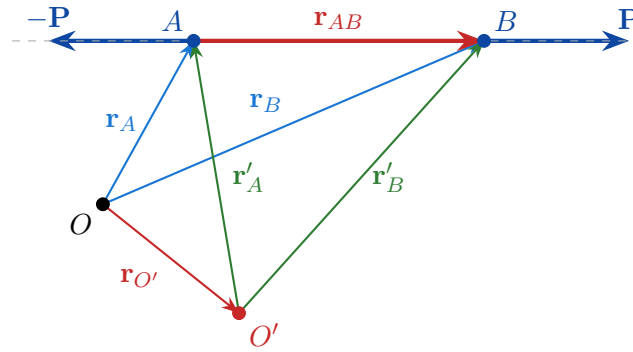
1.4 Theory of Couples & Moments

Theorem / Law: Proof 6: Origin Invariance of Couple Moment

The vector moment of a couple $(-\mathbf{P}, +\mathbf{P})$ acting at $A(\mathbf{r}_A)$ and $B(\mathbf{r}_B)$ is an invariant free vector independent of reference origin O :

$$\mathbf{G}_O = \mathbf{r}_{AB} \times \mathbf{P} \quad (1.4)$$

Rigorous Proof / Derivation: Derivation of Couple Moment Invariance



Moment about origin O : $\mathbf{G}_O = \mathbf{r}_A \times (-\mathbf{P}) + \mathbf{r}_B \times \mathbf{P} = (\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{P} = \mathbf{r}_{AB} \times \mathbf{P}$.

Moment about shifted origin O' ($\mathbf{r}'_A = \mathbf{r}_A - \mathbf{r}_{O'}$, $\mathbf{r}'_B = \mathbf{r}_B - \mathbf{r}_{O'}$):

$$\mathbf{G}_{O'} = (\mathbf{r}_A - \mathbf{r}_{O'}) \times (-\mathbf{P}) + (\mathbf{r}_B - \mathbf{r}_{O'}) \times \mathbf{P} = (\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{P} = \mathbf{G}_O$$

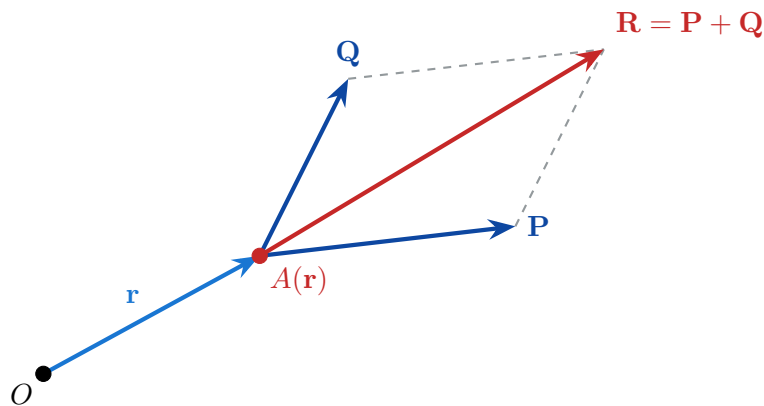
Thus, the moment of a couple is completely invariant under shift of origin.

Theorem / Law: Proof 7: Varignon's Theorem of Moments

The algebraic sum of vector moments of two coplanar forces $\mathbf{P}$ and $\mathbf{Q}$ about any point O equals the moment of their resultant $\mathbf{R} = \mathbf{P} + \mathbf{Q}$ about O :

$$\mathbf{M}_O(\mathbf{R}) = \mathbf{M}_O(\mathbf{P}) + \mathbf{M}_O(\mathbf{Q}) \quad (1.5)$$

Rigorous Proof / Derivation: Derivation of Varignon's Theorem



Taking vector cross product of position vector $\mathbf{r}$ with resultant $\mathbf{R} = \mathbf{P} + \mathbf{Q}$:

$$\mathbf{r} \times \mathbf{R} = \mathbf{r} \times (\mathbf{P} + \mathbf{Q}) = (\mathbf{r} \times \mathbf{P}) + (\mathbf{r} \times \mathbf{Q}) \implies \mathbf{M}_O(\mathbf{R}) = \mathbf{M}_O(\mathbf{P}) + \mathbf{M}_O(\mathbf{Q})$$

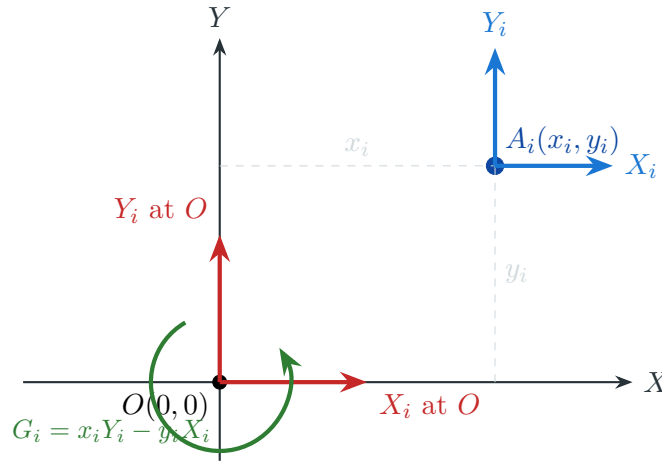
1.5 Reduction of Coplanar System & Line of Action

Theorem / Law: Proof 8 & 9: Reduction of Coplanar Forces and Line of Action

1. **Reduction:** Any system of coplanar forces (X_i, Y_i) acting at (x_i, y_i) reduces at origin O to a single force (X, Y) and a couple $G = \sum(x_i Y_i - y_i X_i)$. 2. **Line of Action:** The equation of the line of action of resultant $R = \sqrt{X^2 + Y^2}$ is:

$$xY - yX = G \quad (1.6)$$

Rigorous Proof / Derivation: Derivation of Reduction and Line of Action



Transferring force (X_i, Y_i) at $A_i(x_i, y_i)$ to origin O introduces an equivalent force (X_i, Y_i) at O plus a couple moment $G_i = x_i Y_i - y_i X_i$. Summing over all forces yields $X = \sum X_i, Y = \sum Y_i, G = \sum(x_i Y_i - y_i X_i)$.

Shift base point from $O(0,0)$ to $O'(\xi, \eta)$. New couple moment $G' = G - \xi Y + \eta X$. For $O'(\xi, \eta)$ to lie on the line of action of resultant force R , couple moment must vanish ($G' = 0$):

$$G - \xi Y + \eta X = 0 \implies \xi Y - \eta X = G \implies xY - yX = G$$

Chapter 2

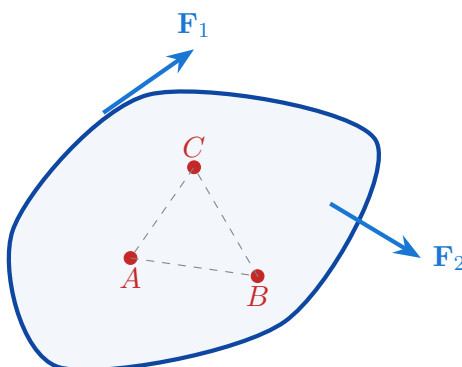
Equilibrium of Forces in 2D & Principle of Virtual Work

2.1 Analytical Conditions of Equilibrium

Theorem / Law: Proof 10 & 11: Conditions of Equilibrium & 3-Point Moment Form

1. **Standard Form:** Coplanar forces are in equilibrium if and only if $X = \sum X_i = 0, Y = \sum Y_i = 0, G = \sum M_O = 0$. 2. **Three-Point Form:** For non-collinear points A, B, C , equilibrium holds iff $M_A = 0, M_B = 0, M_C = 0$.

Rigorous Proof / Derivation: Derivation of 2D Equilibrium Conditions



Resultant force magnitude $R = \sqrt{X^2 + Y^2}$ and couple G . For complete equilibrium, no translational motion requires $R = 0 \implies X = 0, Y = 0$, and no rotational motion requires $G = 0$. If $M_A = 0$, resultant R must either be zero or pass through A . If $M_B = 0$, R must pass through B . Thus line of action of R is line AB . If $M_C = 0$ where C is not on line AB , R cannot pass through C unless $R = 0$. Hence $R = 0$ and $G = 0$.

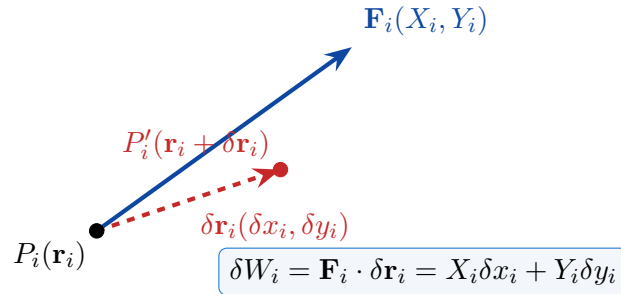
2.2 Principle of Virtual Work

Theorem / Law: Proof 12: Principle of Virtual Work

A coplanar system of rigid bodies is in static equilibrium if and only if the net virtual work performed by all active forces vanishes for every small arbitrary virtual displacement:

$$\sum \delta W = \sum_{i=1}^n (X_i \delta x_i + Y_i \delta y_i) = 0 \quad (2.1)$$

Rigorous Proof / Derivation: Derivation of Virtual Work Principle

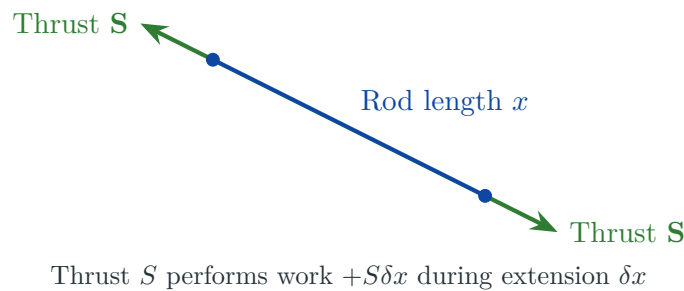


For a particle P_i at (x_i, y_i) under force $\mathbf{F}_i(X_i, Y_i)$, giving a virtual displacement $\delta \mathbf{r}_i(\delta x_i, \delta y_i)$, virtual work is $\delta W_i = \mathbf{F}_i \cdot \delta \mathbf{r}_i = X_i \delta x_i + Y_i \delta y_i$. Summing over system: $\sum \delta W_i = \sum (X_i \delta x_i + Y_i \delta y_i)$. If in equilibrium, $\mathbf{F}_{\text{net}} = \mathbf{0} \implies \sum \delta W = 0$.

Theorem / Law: Proof 13 & 14: Constraint Forces and Active Work Expressions

1. **Constraints:** Reactions at smooth surfaces, fixed hinges, inextensible strings, and rolling without slipping do **zero** net virtual work. 2. **Active Work:** Gravity W : $\delta W = -W \delta y_{cg}$; Elastic String T : $\delta W = -T \delta l$; Rod Thrust S : $\delta W = +S \delta x$.

Rigorous Proof / Derivation: Derivation of Active Force Work Terms



For gravity, potential energy is $V = W y_{cg} \implies \delta W = -\delta V = -W \delta y_{cg}$.

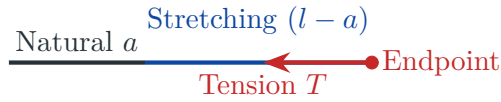
For a tension T pulling two endpoints inwards separated by distance l , virtual work for extension δl is $\delta W = -T \delta l$.

For thrust S pushing endpoints outwards separated by length x , virtual work for extension δx is $\delta W = +S \delta x$.

Theorem / Law: Proof 15: Application of Hooke's Law

Tension $T = \lambda \frac{l-a}{a}$ in elastic string of natural length a stretched to l . Substituting into virtual work yields:

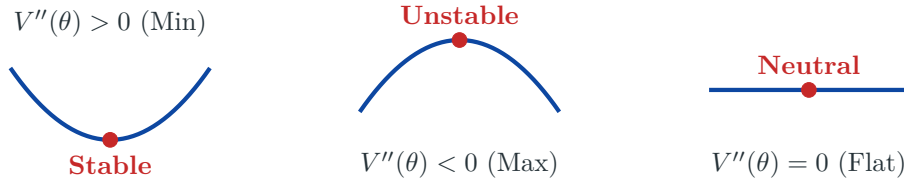
$$\delta W_{\text{elastic}} = -T\delta l = -\lambda \left(\frac{l-a}{a} \right) \delta l \quad (2.2)$$

Rigorous Proof / Derivation: Derivation of Hooke's Law Virtual Work

Hooke's Law states tension is proportional to extension per unit natural length: $T = \lambda \left(\frac{l-a}{a} \right)$. Work done by tension during small extension δl is $\delta W = -T\delta l = -\lambda \left(\frac{l-a}{a} \right) \delta l$.

2.3 Stability of Equilibrium**Theorem / Law: Proof 16: Stability Criteria of Equilibrium**

For conservative system with potential energy $V(\theta)$: Equilibrium holds if $V'(\theta) = 0$. 1. ****Stable****: $V''(\theta) > 0$ (PE is a minimum). 2. ****Unstable****: $V''(\theta) < 0$ (PE is a maximum). 3. ****Neutral****: $V''(\theta) = 0$.

Rigorous Proof / Derivation: Derivation of Stability Criteria

Expanding $V(\theta)$ about equilibrium position θ_0 using Taylor series ($\theta = \theta_0 + x$):

$$V(\theta) = V(\theta_0) + xV'(\theta_0) + \frac{x^2}{2}V''(\theta_0) + \dots$$

Since $V'(\theta_0) = 0$, $\Delta V = V(\theta) - V(\theta_0) \approx \frac{1}{2}x^2V''(\theta_0)$. If $V''(\theta_0) > 0$, $\Delta V > 0$ (potential energy increases upon displacement), creating a restoring force towards θ_0 (Stable). If $V''(\theta_0) < 0$, $\Delta V < 0$, creating a driving force away from θ_0 (Unstable).

Chapter 3

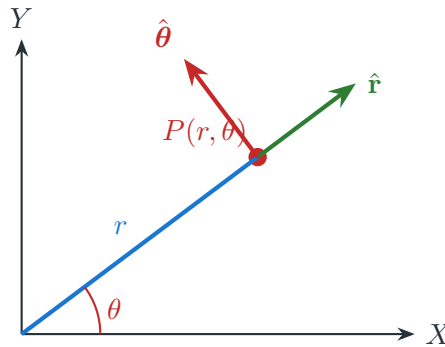
Kinematics & Dynamics of a Particle in 2D (SHM)

3.1 Polar Coordinates Kinematics

Theorem / Law: Proof 17 & 18: Radial and Transverse Velocity and Acceleration

1. **Velocity Components:** Radial $v_r = \dot{r}$, Transverse $v_\theta = r\dot{\theta}$. 2. **Acceleration Components:** Radial $a_r = \ddot{r} - r\dot{\theta}^2$, Transverse $a_\theta = \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta}) = 2\dot{r}\dot{\theta} + r\ddot{\theta}$.

Rigorous Proof / Derivation: Derivation of Polar Velocity and Acceleration



Unit vectors: $\hat{\mathbf{r}} = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$, $\hat{\boldsymbol{\theta}} = -\sin \theta \mathbf{i} + \cos \theta \mathbf{j} \implies \dot{\hat{\mathbf{r}}} = \dot{\theta} \hat{\boldsymbol{\theta}}$, $\dot{\hat{\boldsymbol{\theta}}} = -\dot{\theta} \hat{\mathbf{r}}$.

Position vector $\mathbf{r} = r\hat{\mathbf{r}}$. Differentiating wrt t :

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{r}\hat{\mathbf{r}} + r\dot{\hat{\mathbf{r}}} = \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}} \implies v_r = \dot{r}, \quad v_\theta = r\dot{\theta}$$

Differentiating velocity vector $\mathbf{v}$ wrt t :

$$\begin{aligned} \mathbf{a} &= \frac{d\mathbf{v}}{dt} = \ddot{r}\hat{\mathbf{r}} + \dot{r}(\dot{\theta}\hat{\boldsymbol{\theta}}) + (\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\boldsymbol{\theta}} + r\dot{\theta}(-\dot{\theta}\hat{\mathbf{r}}) \\ &= (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\boldsymbol{\theta}} \implies a_r = \ddot{r} - r\dot{\theta}^2, \quad a_\theta = \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta}) \end{aligned}$$

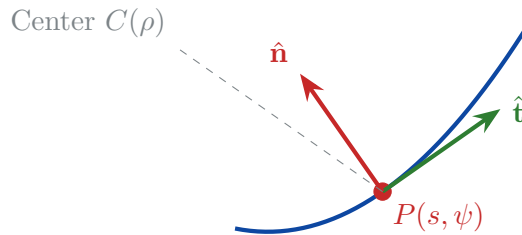
3.2 Intrinsic Coordinates Kinematics

Theorem / Law: Proof 19: Tangential and Normal Acceleration

Along arc $s(\psi)$ with principal normal $\hat{\mathbf{n}}$ towards center of curvature C :

$$a_t = \frac{dv}{dt} = v \frac{dv}{ds} = \ddot{s}, \quad a_n = \frac{v^2}{\rho} \quad (3.1)$$

Rigorous Proof / Derivation: Derivation of Tangential and Normal Acceleration



Velocity vector $\mathbf{v} = v\hat{\mathbf{t}} = \dot{s}\hat{\mathbf{t}}$. Acceleration $\mathbf{a} = \frac{d}{dt}(v\hat{\mathbf{t}}) = \dot{v}\hat{\mathbf{t}} + v\frac{d\hat{\mathbf{t}}}{dt}$.

Using chain rule $\frac{d\hat{\mathbf{t}}}{dt} = \frac{d\hat{\mathbf{t}}}{d\psi} \frac{d\psi}{ds} \frac{ds}{dt} = (\hat{\mathbf{n}}) \left(\frac{1}{\rho}\right) (v) = \frac{v}{\rho}\hat{\mathbf{n}}$:

$$\mathbf{a} = \dot{v}\hat{\mathbf{t}} + \frac{v^2}{\rho}\hat{\mathbf{n}} \implies a_t = \dot{v} = v \frac{dv}{ds} = \ddot{s}, \quad a_n = \frac{v^2}{\rho}$$

3.3 Angular Momentum & Rectilinear SHM

Theorem / Law: Proof 20: Conservation of Angular Momentum under Central Force

For central force $\mathbf{F} = F(r)\hat{\mathbf{r}}$, angular momentum per unit mass $h = r^2\dot{\theta}$ is constant.

Rigorous Proof / Derivation: Derivation of Angular Momentum Conservation

Transverse equation of motion under purely radial central force ($F_\theta = 0$):

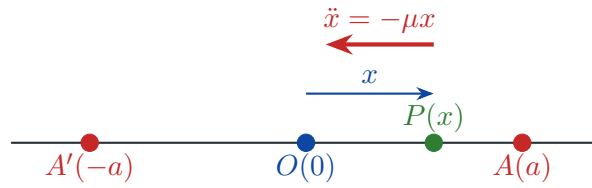
$$a_\theta = \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta}) = 0 \implies \frac{d}{dt}(r^2\dot{\theta}) = 0 \implies r^2\dot{\theta} = h = \text{constant}$$

Areal velocity $\frac{dA}{dt} = \frac{1}{2}r^2\dot{\theta} = \frac{h}{2} = \text{constant}$. Angular momentum $L = mr^2\dot{\theta} = mh$ is strictly conserved.

Theorem / Law: Proof 21, 22 & 23: Velocity, Displacement, and Period in Rectilinear SHM

For equation $\ddot{x} = -\mu x$: 1. **Velocity:** $v^2 = \mu(a^2 - x^2)$. 2. **Displacement:** $x(t) = a \cos(\sqrt{\mu}t)$ or $x(t) = a \sin(\sqrt{\mu}t)$. 3. **Period & Frequency:** $T = \frac{2\pi}{\sqrt{\mu}}, n = \frac{\sqrt{\mu}}{2\pi}$.

Rigorous Proof / Derivation: Derivation of SHM Formulae



Integrating $v \frac{dv}{dx} = -\mu x \implies \frac{v^2}{2} = -\mu \frac{x^2}{2} + C$. Using boundary condition $v = 0$ at $x = a \implies C = \frac{\mu a^2}{2} \implies v^2 = \mu(a^2 - x^2)$.

Integrating $\frac{dx}{\sqrt{a^2 - x^2}} = -\sqrt{\mu} dt \implies \cos^{-1}(x/a) = \sqrt{\mu} t \implies x(t) = a \cos(\sqrt{\mu} t)$.

Time taken for one full cycle $A \rightarrow A' \rightarrow A$: $\Delta(\sqrt{\mu} t) = 2\pi \implies T = \frac{2\pi}{\sqrt{\mu}}, n = 1/T = \frac{\sqrt{\mu}}{2\pi}$.

Chapter 4

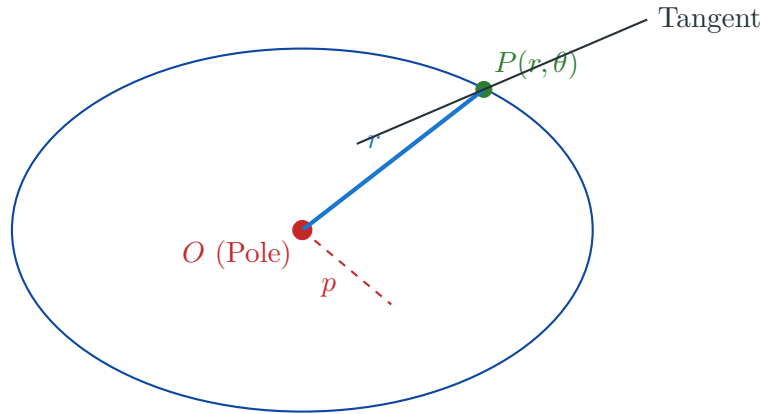
Central Orbits, Kepler's Laws & Constrained Motion

4.1 Differential Equations of Central Orbits

Theorem / Law: Proof 24 & 25: Polar and Pedal Equations of Central Orbits

1. **Polar Form** ($u = 1/r$): $\frac{d^2u}{d\theta^2} + u = \frac{F(u)}{h^2u^2}$. 2. **Pedal Form** ($p = r \sin \phi$): $\frac{h^2}{p^3} \frac{dp}{dr} = F(r)$.

Rigorous Proof / Derivation: Derivation of Polar and Pedal Orbit Equations



Radial equation: $\ddot{r} - r\dot{\theta}^2 = -F$. Substituting $r = 1/u$ and $\dot{\theta} = hu^2$:

$$\dot{r} = \frac{d}{d\theta} \left(\frac{1}{u} \right) \dot{\theta} = -\frac{1}{u^2} \frac{du}{d\theta} (hu^2) = -h \frac{du}{d\theta} \implies \ddot{r} = -h \frac{d^2u}{d\theta^2} \dot{\theta} = -h^2 u^2 \frac{d^2u}{d\theta^2}$$

Substituting into radial equation: $-h^2 u^2 \frac{d^2u}{d\theta^2} - \frac{1}{u} (h^2 u^4) = -F \implies \frac{d^2u}{d\theta^2} + u = \frac{F}{h^2 u^2}$.

For pedal form, $1/p^2 = u^2 + (du/d\theta)^2$. Differentiating wrt θ : $-\frac{2}{p^3} \frac{dp}{d\theta} = 2u \frac{du}{d\theta} + 2 \frac{du}{d\theta} \frac{d^2u}{d\theta^2} = 2 \frac{du}{d\theta} \left(u + \frac{d^2u}{d\theta^2} \right) = 2 \frac{du}{d\theta} \frac{F}{h^2 u^2}$. Since $\frac{dp}{d\theta} = \frac{dp}{dr} \frac{dr}{d\theta} = \frac{dp}{dr} \left(-\frac{1}{u^2} \frac{du}{d\theta} \right)$, simplifying yields $\frac{h^2}{p^3} \frac{dp}{dr} = F(r)$.

Theorem / Law: Proof 26: Velocity at Any Point of Central Orbit

$$v^2 = h^2 \left[u^2 + \left(\frac{du}{d\theta} \right)^2 \right] = \frac{h^2}{p^2} \quad (4.1)$$

Rigorous Proof / Derivation: Derivation of Orbit Velocity Formula

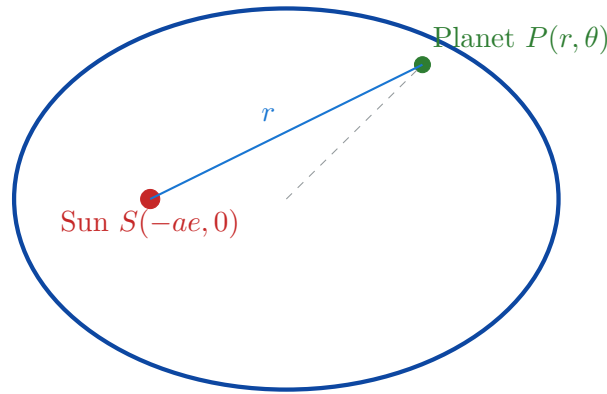
Radial velocity $v_r = \dot{r} = -h \frac{du}{d\theta}$, transverse velocity $v_\theta = r\dot{\theta} = hu$.

Total speed squared: $v^2 = v_r^2 + v_\theta^2 = h^2 \left[u^2 + \left(\frac{du}{d\theta} \right)^2 \right]$.

Using pedal relation $p = r \sin \phi \implies v = \frac{r\dot{\theta}}{\sin \phi} = \frac{h}{p} \implies v^2 = \frac{h^2}{p^2}$.

4.2 Inverse Square Law & Kepler's Laws**Theorem / Law: Proof 27, 28, 29 & 30: Conic Orbits and Kepler's Three Laws**

Under central force $F = \mu/r^2$: 1. **First Law:** Orbit is conic $l/r = 1 + e \cos \theta$ ($l = h^2/\mu$). Ellipse for $e < 1$ ($v^2 = \mu(2/r - 1/a)$). 2. **Second Law:** Areal velocity is constant ($\frac{dA}{dt} = \frac{h}{2}$). 3. **Third Law:** Period squared $T^2 = \frac{4\pi^2}{\mu} a^3 \implies T^2 \propto a^3$.

Rigorous Proof / Derivation: Derivation of Inverse Square Law & Kepler's Laws

For $F = \mu u^2$, polar equation becomes $\frac{d^2 u}{d\theta^2} + u = \frac{\mu}{h^2}$. General solution $u = \frac{\mu}{h^2} [1 + e \cos(\theta - \theta_0)] \implies \frac{l}{r} = 1 + e \cos \theta$ where $l = h^2/\mu$. This proves Kepler's 1st Law (Elliptic orbit with Sun at focus for $e < 1$).

Kepler's 2nd Law follows directly from transverse equation $r^2 \dot{\theta} = h \implies \frac{dA}{dt} = \frac{h}{2} = \text{const.}$

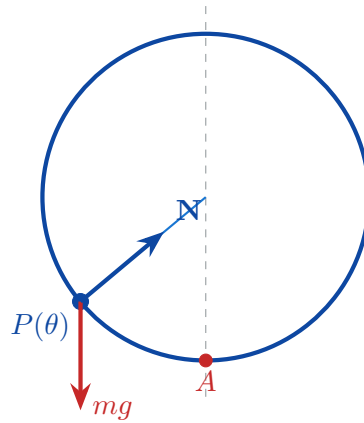
For Kepler's 3rd Law, total area of ellipse is πab . Period $T = \frac{\text{Total Area}}{\text{Areal Velocity}} = \frac{\pi ab}{h/2} = \frac{2\pi ab}{h}$. Since $b^2 = al = a(h^2/\mu) \implies h = \sqrt{\mu l} = \frac{b\sqrt{\mu}}{\sqrt{a}}$:

$$T = \frac{2\pi ab}{b\sqrt{\mu}/\sqrt{a}} = \frac{2\pi a^{3/2}}{\sqrt{\mu}} \implies T^2 = \left(\frac{4\pi^2}{\mu} \right) a^3 \implies T^2 \propto a^3$$

4.3 Constrained Motion on Vertical Curves**Theorem / Law: Proof 31 & 32: Motion on Smooth Vertical Circle**

1. **Velocity & Reaction:** $v^2 = u^2 - 2ga(1 - \cos \theta)$, $N = \frac{m}{a} [u^2 - 2ga + 3ga \cos \theta]$. Loop complete if $u \geq \sqrt{5ga}$. 2. **Small Oscillations:** $\ddot{s} \approx -\left(\frac{g}{a}\right)s \implies T = 2\pi \sqrt{\frac{a}{g}}$.

Rigorous Proof / Derivation: Derivation of Vertical Circle Motion



Tangential equation $mv \frac{dv}{ds} = -mg \sin \theta$. With $s = a\theta$, $v \frac{dv}{d\theta} = -ga \sin \theta$. Integrating from $\theta = 0 (v = u)$: $v^2 = u^2 - 2ga(1 - \cos \theta)$.

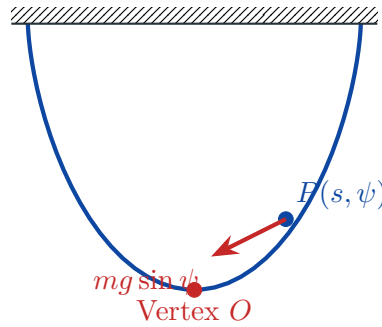
Normal equation $\frac{mv^2}{a} = N - mg \cos \theta \implies N = mg \cos \theta + \frac{m}{a}[u^2 - 2ga(1 - \cos \theta)] = \frac{m}{a}[u^2 - 2ga + 3ga \cos \theta]$.

For small θ , $\sin \theta \approx \theta = s/a \implies \ddot{s} = -g(s/a) \implies T = 2\pi\sqrt{a/g}$.

Theorem / Law: Proof 33 & 34: Motion on Vertical Cycloid (Isochronous SHM)

For cycloid $s = 4a \sin \psi$: $\ddot{s} = -g \sin \psi = -\left(\frac{g}{4a}\right)s$. Exact isochronous SHM with $T = 4\pi\sqrt{\frac{a}{g}}$ and fall time to vertex $t_{\text{vertex}} = \pi\sqrt{\frac{a}{g}}$.

Rigorous Proof / Derivation: Derivation of Cycloidal SHM



Equation of motion along cycloid arc s : $\ddot{s} = -g \sin \psi$. Substituting intrinsic cycloid formula $\sin \psi = s/(4a)$:

$$\ddot{s} = -\left(\frac{g}{4a}\right)s \implies \text{Exact SHM } \ddot{s} = -\omega^2 s \quad \text{where } \omega = \sqrt{\frac{g}{4a}} = \frac{1}{2}\sqrt{\frac{g}{a}}$$

Time period $T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{1}{2}\sqrt{g/a}} = 4\pi\sqrt{\frac{a}{g}}$ (Isochronous, independent of amplitude s_0).

Integrating $\dot{s}^2 = \omega^2(s_0^2 - s^2) \implies t = \frac{1}{\omega} \cos^{-1}(s/s_0) = 2\sqrt{\frac{a}{g}} \cos^{-1}\left(\frac{\sin \psi}{\sin \psi_0}\right)$. Fall to vertex ($s = 0$) takes $t_{\text{vertex}} = \frac{\pi}{2\omega} = \pi\sqrt{\frac{a}{g}}$.

Chapter 5

Forces in 3D, Dyname, Poinot's Central Axis & Moments of Inertia

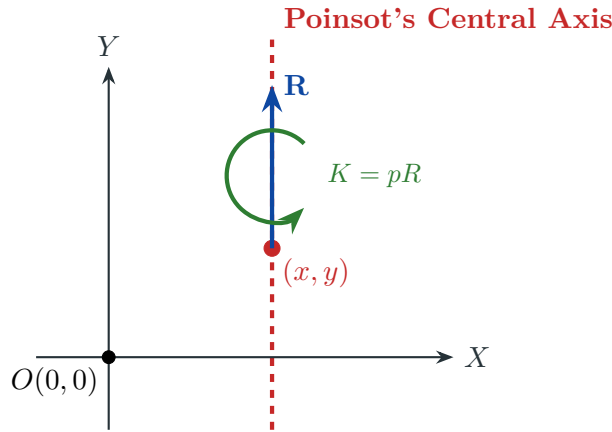
5.1 Reduction of 3D Forces & Poinot's Central Axis

Theorem / Law: Proof 35, 36 & 37: 3D Force Reduction, Pitch, and Poinot's Central Axis

1. **Reduction:** 3D forces reduce at O to force $\mathbf{R}(X, Y, Z)$ and couple $\mathbf{G}(L, M, N)$. 2. **Pitch:** $p = \frac{\mathbf{R} \cdot \mathbf{G}}{R^2} = \frac{XL + YM + ZN}{X^2 + Y^2 + Z^2}$. 3. **Central Axis:** Equation of Poinot's Central Axis:

$$\frac{L - (yZ - zY)}{X} = \frac{M - (zX - xZ)}{Y} = \frac{N - (xY - yX)}{Z} = p \quad (5.1)$$

Rigorous Proof / Derivation: Derivation of Poinot's Central Axis



Forces $\mathbf{F}_i(X_i, Y_i, Z_i)$ at (x_i, y_i, z_i) reduce at origin O to $\mathbf{R} = (\sum X_i, \sum Y_i, \sum Z_i)$ and couple $\mathbf{G} = (L, M, N)$ where $L = \sum (y_i Z_i - z_i Y_i)$, $M = \sum (z_i X_i - x_i Z_i)$, $N = \sum (x_i Y_i - y_i X_i)$.

Shifting origin to $O'(x, y, z)$, new couple components are $L' = L - (yZ - zY)$, $M' = M - (zX - xZ)$, $N' = N - (xY - yX)$.

For $O'(x, y, z)$ to lie on Central Axis, couple vector $\mathbf{G}'$ must be parallel to force $\mathbf{R}$ ($\mathbf{G}' = p\mathbf{R}$):

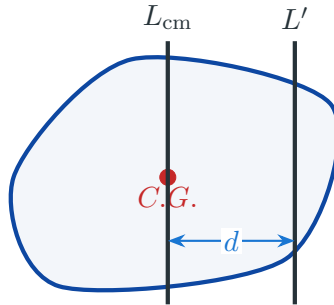
$$\frac{L'}{X} = \frac{M'}{Y} = \frac{N'}{Z} = p \implies \frac{L - (yZ - zY)}{X} = \frac{M - (zX - xZ)}{Y} = \frac{N - (xY - yX)}{Z} = p$$

5.2 Moments of Inertia Theorems & Standard Bodies

Theorem / Law: Proof 38 & 39: Parallel and Perpendicular Axes Theorems

1. **Parallel Axes Theorem:** $I_{L'} = I_{cm} + Md^2$. 2. **Perpendicular Axes Theorem (Plane Lamina):** $I_z = I_x + I_y$.

Rigorous Proof / Derivation: Derivation of M.I. Theorems



For Parallel Axes Theorem, let mass element m_i be at distance r_i from C.G. axis L_{cm} . Distance from parallel axis L' is $r'_i = r_i + d$:

$$I_{L'} = \sum m_i(r_i + d)^2 = \sum m_i r_i^2 + 2d \sum m_i r_i + d^2 \sum m_i = I_{cm} + 0 + Md^2 = I_{cm} + Md^2$$

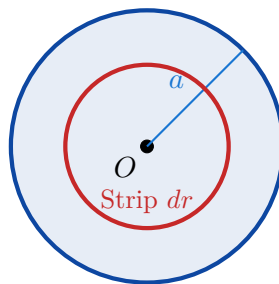
For Perpendicular Axes Theorem of plane lamina in XY plane ($z_i = 0$):

$$I_x + I_y = \sum m_i y_i^2 + \sum m_i x_i^2 = \sum m_i (x_i^2 + y_i^2) = I_z$$

Theorem / Law: Proof 40: Standard Moments of Inertia Derivations

Integrations for symmetric bodies: Thin Rod ($I = \frac{1}{3}ma^2$), Circular Disc ($I = \frac{1}{2}ma^2$), Solid Sphere ($I = \frac{2}{5}ma^2$).

Rigorous Proof / Derivation: Derivation of Standard M.I. Formulae



For a circular disc of mass m and radius a , surface density $\sigma = \frac{m}{\pi a^2}$. Consider a thin concentric ring of radius r and width dr . Mass of ring $dm = \sigma(2\pi r dr) = \frac{2m}{a^2} r dr$. Moment of inertia of ring about central axis $dI = dm \cdot r^2 = \frac{2m}{a^2} r^3 dr$. Integrating from $r = 0$ to $r = a$:

$$I_{axis} = \int_0^a \frac{2m}{a^2} r^3 dr = \frac{2m}{a^2} \left[\frac{r^4}{4} \right]_0^a = \frac{1}{2}ma^2$$

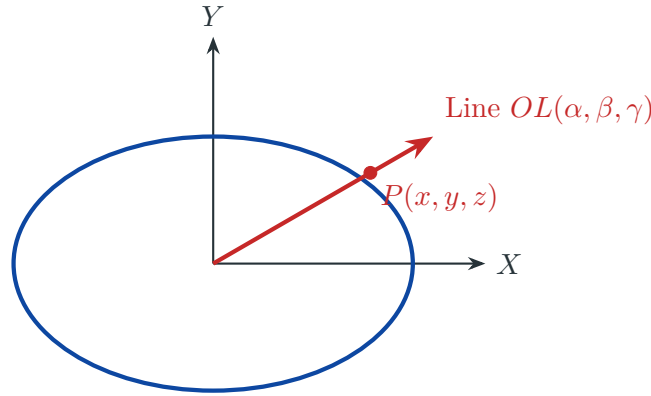
Using Perpendicular Axes Theorem $I_z = I_x + I_y = 2I_{dia} \implies I_{dia} = \frac{1}{2}I_{axis} = \frac{1}{4}ma^2$.

5.3 Inclined Lines & Momental Ellipsoid

Theorem / Law: Proof 41 & 42: M.I. About Inclined Line & Momental Ellipsoid

1. **Inclined Line** (α, β, γ) : $I = I_{xx}\alpha^2 + I_{yy}\beta^2 + I_{zz}\gamma^2 - 2I_{xy}\alpha\beta - 2I_{yz}\beta\gamma - 2I_{zx}\gamma\alpha$.
2. **Momental Ellipsoid**: Quadric surface $I_{xx}x^2 + I_{yy}y^2 + I_{zz}z^2 - 2I_{xy}xy - 2I_{yz}yz - 2I_{zx}zx = 1$.

Rigorous Proof / Derivation: Derivation of Inclined Line M.I. and Momental Ellipsoid



Perpendicular distance p_i of mass $m_i(x_i, y_i, z_i)$ from line $OL(\alpha, \beta, \gamma)$ satisfies $p_i^2 = (x_i^2 + y_i^2 + z_i^2)(\alpha^2 + \beta^2 + \gamma^2) - (\alpha x_i + \beta y_i + \gamma z_i)^2$. Summing over m_i :

$$I_{OL} = I_{xx}\alpha^2 + I_{yy}\beta^2 + I_{zz}\gamma^2 - 2I_{xy}\alpha\beta - 2I_{yz}\beta\gamma - 2I_{zx}\gamma\alpha$$

Set point $P(x, y, z)$ on line OL at distance $r = 1/\sqrt{I_{OL}}$ from origin O . Then $x = r\alpha, y = r\beta, z = r\gamma \implies \alpha = x/r, \beta = y/r, \gamma = z/r$. Substituting into I_{OL} :

$$I_{OL} = I_{xx} \left(\frac{x}{r}\right)^2 + I_{yy} \left(\frac{y}{r}\right)^2 + I_{zz} \left(\frac{z}{r}\right)^2 - 2I_{xy} \frac{xy}{r^2} - 2I_{yz} \frac{yz}{r^2} - 2I_{zx} \frac{zx}{r^2}$$

Multiplying by $r^2 = 1/I_{OL}$ yields the equation of the **Moment of Inertia Ellipsoid**:

$$I_{xx}x^2 + I_{yy}y^2 + I_{zz}z^2 - 2I_{xy}xy - 2I_{yz}yz - 2I_{zx}zx = 1$$